

Lecture 7

In Class Exercises



Exercise 1

- Write an iterative version of RANDOMIZED-SELECT

Exercise 1 (solution)

ITERATIVE-RANDOMIZED-SELECT(A, p, r, i)

while $p < r$

$q = \text{RANDOMIZED-PARTITION}(A, p, r)$

$k = q - p + 1$

if $i == k$

return $A[q]$

elseif $i < k$

$r = q - 1$

else $p = q + 1$

$i = i - k$

return $A[p]$

Exercise 2

- Suppose that RANDOMIZED-SELECT is used to select the minimum element of the array $A = \{3, 2, 9, 0, 7, 5, 4, 8, 6, 1\}$. Describe a sequence of partitions that results in a worst-case performance of RANDOMIZED-SELECT.

Exercise 2 (solution)

- First 9 is selected as pivot, then 8, then 7 ... all the way until the minimum element is found. With this kind of partitioning, we always recurse on a subarray which is only one element smaller than the previous subarray. Example below (without the position exchanges, just for illustration):

pivot **9** {3, 2, 0, 7, 5, 4, 8, 6, 1}

pivot **8** {3, 2, 0, 7, 5, 4, 6, 1}

pivot **7** {3, 2, 0, 5, 4, 6, 1}

pivot **6** {3, 2, 0, 5, 4, 1}

pivot **5** {3, 2, 0, 4, 1}

pivot **4** {3, 2, 0, 1}

pivot **3** {2, 0, 1}

pivot **2** {0, 1}

pivot **1** {0}